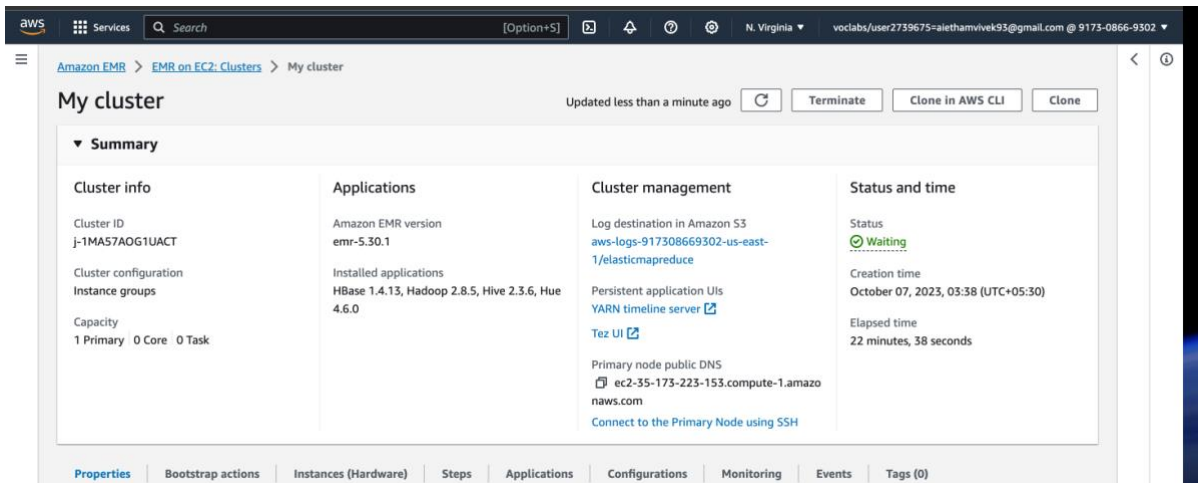


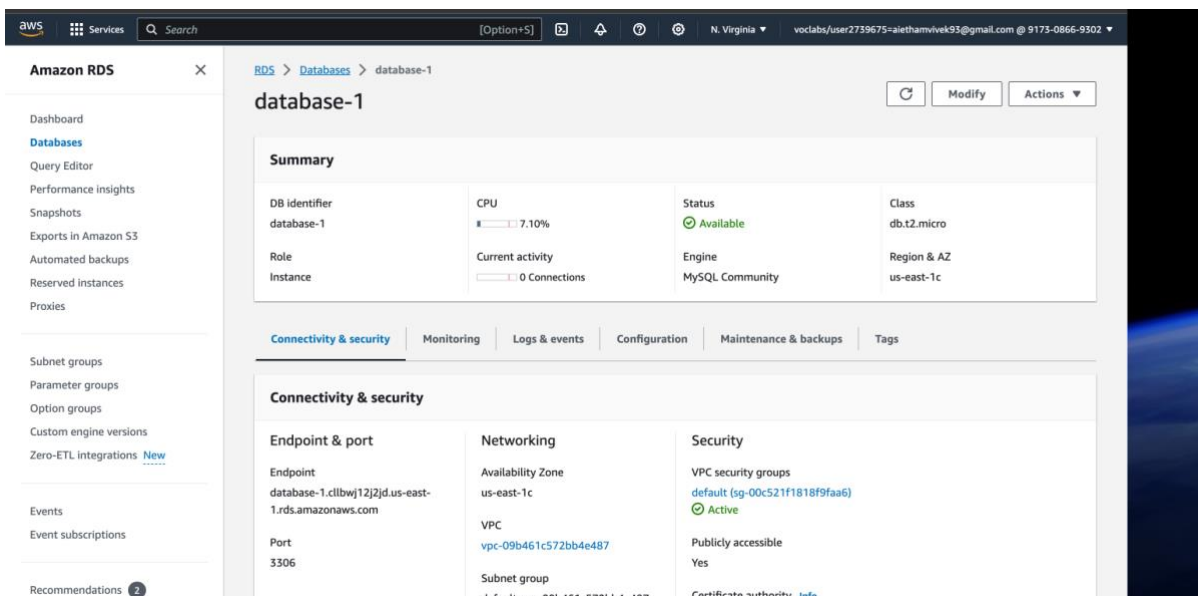
Task – 1

EMR- RDS Connection and importing tables.

Step 1. Setting up EMR Cluster.



Ste2. Creating RDS mysql data base.



Step 3. Connecting to the cluster from Local machine using .pem file and command

```
ssh -i Upgrad.pem hadoop@ec2-35-173-223-153.compute-1.amazonaws.com
```

```
Aiethams-MacBook-Air:Upgrad aiethamvivek$ ssh -i Upgrad.pem hadoop@ec2-35-173-223-153.compute-1.amazonaws.com
The authenticity of host 'ec2-35-173-223-153.compute-1.amazonaws.com (35.173.223.153)' can't be established.
ED25519 key fingerprint is SHA256:fb6shVmV/Cg9xihvi18cE6t5zs1NRK67py+jDTA109k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-35-173-223-153.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Last login: Fri Oct 6 22:18:50 2023
```

```
  _ |  _ |  _ )
 _ |  (  /   Amazon Linux 2 AMI
---| \---|---|
```

```
https://aws.amazon.com/amazon-linux-2/
92 package(s) needed for security, out of 160 available
Run "sudo yum update" to apply all updates.
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
```

```
EEEEEEEEEEEEEEEEEEEE MMMMMMMM MMMMMMMM RRRRRRRRRRRRRRRR
E::::::::::::::::::::E M::::::::M M::::::::M R::::::::::::R
EE::::::::EEEEEEEEEEE M::::::::M M::::::::M R::::RRRRRR::::R
 E::::E EEEEE M::::::::M M::::::::M RR:::R R:::R
 E::::E M::::::::M M:::M M:::M R:::R R:::R
 E::::EEEEEEEEEE M:::M M:::M M:::M R::RRRRRR::::R
 E::::::::::::E M:::M M:::M M:::M R::::::::::::RR
 E::::EEEEEEEEEE M:::M M:::M M:::M R::RRRRRR::::R
 E::::E M:::M M:::M M:::M R:::R R:::R
 E::::E EEEEE M:::M MMM M:::M R:::R R:::R
 EE::::::::EEEEEEEEEE M:::M M:::M R:::R R:::R
 E::::::::::::E M:::M M:::M RR:::R R:::R
 EEEEEEEEEEEEEEEEEEE MMMMMMMM MMMMMMMM RRRRRRR RRRRRR
```

Step 4. Downloading the 2 csv files using the commands:

```
wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
```

```
wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
```

```
[hadoop@ip-172-30-0-46 ~]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
--2023-10-06 22:37:47-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 3.5.11.134, 3.5.29.149, 3.5.25.88, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|3.5.11.134|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 914029540 (872M) [text/csv]
Saving to: 'yellow_tripdata_2017-01.csv.1'

100%[=====] 914,029,540 26.4MB/s in 34s

2023-10-06 22:38:21 (25.4 MB/s) - 'yellow_tripdata_2017-01.csv.1' saved [914029540/914029540]

[hadoop@ip-172-30-0-46 ~]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
--2023-10-06 22:38:27-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 54.231.129.241, 52.216.162.3, 52.217.200.57, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|54.231.129.241|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 863487050 (823M) [text/csv]
Saving to: 'yellow_tripdata_2017-02.csv'

100%[=====] 863,487,050 24.1MB/s in 37s

2023-10-06 22:39:05 (22.0 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-30-0-46 ~]$
```

Step 5. Connection to my Sql using command:

```
mysql -h database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
```

```
[hadoop@ip-172-30-0-46 ~]$ mysql -h database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 8.0.33 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> █
```

Step 6. Creating table using command:

```
CREATE TABLE taxi (VendorID integer, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime,
passenger_count integer, trip_distance float, RatecodeID integer, store_and_fwd_flag varchar(1),
PULocationID integer, DOLocationID integer, payment_type integer, fare_amount float, extra float, mta_tax
float, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, Airport_fee
float);
```

```
[hadoop@ip-172-30-0-46 ~]$ mysql -h database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com
ERROR 2009 (HY000): Authentication plugin 'sha256_password' cannot be loaded: /usr/lib64/mysql/plugin/sha256_password.so: cannot open shared object file: No such file or directory
[hadoop@ip-172-30-0-46 ~]$ mysql -h database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 8.0.33 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE yellow;
Query OK, 1 row affected (0.01 sec)

MySQL [(none)]> USE yellow;
Database changed
MySQL [yellow]> CREATE TABLE taxi (( VendorID integer, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count integer, trip_distance float, RatecodeID integer, store_and_fwd_flag varchar(1), PULocationID integer, DOLocationID integer, payment_type integer, fare_amount float, extra float, mta_tax float, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float,Airport_fee float)
-> CREATE TABLE taxi (( VendorID integer, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count integer, trip_distance float, RatecodeID integer, store_and_fwd_flag varchar(1), PULocationID integer, DOLocationID integer, payment_type integer, fare_amount float, extra float, mta_tax float, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float,Airport_fee float);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'VendorID integer, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime,' at line 1
MySQL [yellow]> CREATE TABLE taxi (VendorID integer, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count integer, trip_distance float, RatecodeID integer, store_and_fwd_flag varchar(1), PULocationID integer, DOLocationID integer, payment_type integer, fare_amount float, extra float, mta_tax float, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, Airport_fee float);
Query OK, 0 rows affected (0.03 sec)

MySQL [yellow]> █
```

Step 7. Loading the data from both the file to table 'taxi' that exists in RDS.

```
LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-01.csv' INTO TABLE taxi FIELDS
TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 LINES;
```

```
LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-02.csv' INTO TABLE taxi FIELDS
TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 LINES;
```

```
MySQL [yellow]> LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-01.csv' INTO TABLE taxi FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 LINES;
Query OK, 9718820 rows affected, 65535 warnings (2 min 21.92 sec)
Records: 9718820 Deleted: 0 Skipped: 0 Warnings: 9718820

MySQL [yellow]>
MySQL [yellow]> LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-02.csv' INTO TABLE taxi FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 LINES;
Query OK, 9169775 rows affected, 65535 warnings (2 min 21.84 sec)
Records: 9169775 Deleted: 0 Skipped: 0 Warnings: 9169775

MySQL [yellow]> SELECT count(*) FROM taxi;
+-----+
| count(*) |
+-----+
| 18880595 |
+-----+
1 row in set (43.88 sec)

MySQL [yellow]> █
```